BOARD MEMORANDUM

Evaluation of Two Strategic Investment Opportunities: Recommendation and Capital Allocation

To	Board of Directors, Tiny-Rod Hit Inc.
From	Senior Finance Manager
Date	January 2025
Subject	Project A (emerging-markets expansion) vs. Project B (disruptive technology R&D): NPV/IRR assessment, recommendation, risks and allocation of the US$100M available cash
Decision sought	Approval of Project B and direction on Project A (Section 8)

1. Executive Summary

Recommendation: approve Project B (disruptive technology R&D, US$40M), released in three gated tranches. Do not commit the full US$50M to Project A at this time. Direct management to return within one planning cycle with a scoped, lower-capital market-entry option and a validated business case.

If the Board wishes to pursue both ventures, we recommend the staged allocation in Section 7: US$45M committed at approval, US$45M released only at defined gates, and a US$10M reserve held back for contingencies. Both projects clear the 9% WACC on our directional estimates; the case for Project B is stronger.

Why Project B. On directional estimates, Project B's probability-weighted NPV is about US$149M (11% discount rate), roughly three times the US$51M base-case NPV of Project A (12%). Project B requires US$40M rather than US$50M, and its expected IRR of about 25% (about 29% if successful) compares with about 18% for Project A. It is also built on capabilities the company already has: a strong internal R&D team and deep knowledge of its core market. The 30% chance of failure is material, but it is bounded and can be reduced. With a stage gate after the first US$20M, the loss in the failure case falls from about US$40M to about US$20M.

Why not Project A now. Project A creates value on our base case, but the margin is thinner. Its IRR of about 18% sits only moderately above the 12% risk-adjusted hurdle we apply to emerging markets, and in the bear case its NPV turns negative (–US$19M). Most of its risks are external and hard to control: geopolitical events, regulatory change, currency movements and local competition. The company also has limited prior experience in these markets and would need to build infrastructure and partnerships before earning revenue. Diversification is a real strategic benefit, but shareholders can diversify on their own. It adds value only if the returns clear the cost of capital, and on current assumptions the margin is smaller than Project B's. A lower-cost entry route (for example, a joint venture or licensing pilot) would keep the option open at a fraction of the commitment.

Important caveat. The reference brief gives investment amounts, growth rates and a failure probability, but no revenue base, margin or cost data. The figures in this memo are therefore directional. They rest on the explicit assumptions in Section 2 and should be confirmed against detailed business cases before any tranche is released.

Table 1. Summary comparison

Metric (directional)	Project A: Emerging markets	Project B: Disruptive technology R&D
Capital required	US$50M, fully upfront	US$40M over three years, in gated tranches
Discount rate applied	12% (9% WACC + 3pp country and geopolitical premium)	11% (9% WACC + 2pp technology premium) on probability-weighted cash flows
Base-case NPV	About US$51M (range –US$19M bear to US$136M bull)	About US$149M expected (range –US$38M failure to US$229M success)

Metric (directional)	Project A: Emerging markets	Project B: Disruptive technology R&D
IRR	About 18% (bear 9%; bull 25%)	About 25% expected; about 29% if successful
Payback (undiscounted)	About 7.2 years	About 6.7 years if successful; about 7.3 years expected
Share of value in terminal period	About 78%	About 65% (expected)
Principal downside	Stranded infrastructure, FX loss, political disruption	R&D failure (30%); capital write-off of up to US$40M
Strategic fit	Diversification; new customers; long-term growth	Market leadership; core expertise; defends existing franchise
Assessment	Viable, but deferred pending scoping	Recommended, subject to stage gates

2. Basis of Preparation and Key Assumptions

The analysis is a high-level, directional screen. We have used a 15-year explicit horizon, annual year-end cash flows, after-tax unlevered free cash flow and a terminal value on Year-15 cash flow. Year 0 is the approval date (January 2025). All amounts are in US dollars. Assumptions that go beyond the brief are set out below; none is a management forecast.

Table 2. Key modelling assumptions (directional)

Parameter	Project A	Project B	Basis / comment
Investment	US$50M at Year 0	US$20M (Yr 0), US$15M (Yr 1), US$5M (Yr 2); total US$40M	Brief; phasing is our assumption
Revenue	Year 1 about US$40M; +20% p.a. Years 1–5; +5% p.a. thereafter	Launch in Year 3 at US$30M; +50% p.a. for three years after launch; then +30%, +20%, +10%; then +5%	Growth rates per brief ("over 50%" modelled at 50%, a conservative reading)
Cash margin (after tax, after working capital and capex)	Ramps 0% → 14% by Year 5	Ramps 0% (launch year) → 22% by year 5 after launch	Assumed; B higher due to existing distribution and customers
Failure case	Not applicable (failure risk captured in sensitivities)	30% probability; full US$40M spent; no revenue; no salvage value	Brief; conservative (no IP or licensing recovery)
Discount rate	12% (9% WACC + 3pp)	11% (9% WACC + 2pp) on probability-weighted flows	Board WACC of 9% per brief; premiums are our judgement
Terminal growth	3% after Year 15	3% after Year 15	More conservative than the 5% stabilised growth in the brief
Funding	Within the US$100M available cash	Within the US$100M available cash	Brief dates the cash to May 2025; a four-month timing shift changes NPVs by only a few percent

3. Financial Analysis

3.1 Project A: Emerging-markets expansion

Revenue grows from about US$40M in Year 1 to about US$83M in Year 5 and about US$106M in Year 10. Cash flows are negative at approval and then close to breakeven for the first two years while infrastructure and partnerships are established. The steady-state margin of 14% is reached only in Year 5. At a 12% discount rate the base case gives an NPV of about US$51M and an IRR of about 18%, with payback in about 7.2 years. The result depends heavily on the discount rate: at the 9% WACC the NPV is about US$117M, and at 14% it falls to about US$27M. About 78% of the value sits in the terminal period, so the case depends on sustaining growth after Year 5.

The project is most sensitive to margin and to early revenue. Breakeven at 12% needs a steady-state cash margin of about 6.4% or Year 1 revenue of about US$20M. Probability-weighting the bear, base and bull cases (25/50/25) gives about US$55M, close to the base case but with a wide spread: bear –US$19M, bull US$136M.

3.2 Project B: Disruptive technology R&D

If successful, Project B reaches about US$30M of revenue at launch and about US$182M by Year 10, and it generates an NPV of about US$229M at 11% with an IRR of about 29%. If the R&D fails, the company loses the full US$40M. That outcome has an NPV of about –US$38M. Weighting the two outcomes at 70% and 30% gives an expected NPV of about US$149M and an expected IRR of about 25%. Expected payback is about 7.3 years, and about 65% of expected value sits in the terminal period.

Two features make Project B more attractive than its headline failure rate suggests. First, the expected-value result stays positive across the sensitivities we tested. The 30% failure probability would have to rise to about 86% before the expected NPV reaches zero. Second, staging the spend changes the risk profile. If a stage gate after the first US$20M stops the programme on failure, the expected NPV rises slightly to about US$155M, and the maximum loss falls from US$40M to US$20M.

3.3 Comparative results and sensitivity

Table 3 sets out the sensitivities. The brief gives no operating data, so we tested each main driver separately and then combined the adverse ones.

Table 3. NPV sensitivity and stress tests (US$ millions, directional)

Case	Project A NPV (12%)	Project B expected NPV (11%)	Reading
Base case	US$51M	US$149M	B about 3× A
Discount rate at 9% WACC (no premium)	US$117M	US$240M	Ranking unchanged
Higher discount rate (A 14%; B 13%)	US$27M	US$97M	B remains ahead
Growth slower (A 10% p.a.; B 25% p.a. post-launch)	US$16M	US$49M	B only roughly equal to A base
Growth faster (A 25% p.a.; B 60% p.a. post-launch)	US$73M	US$216M	Upside
Lower steady-state margin (A 8%; B 12%)	US$11M	US$72M	B still ahead
Lower early revenue (A Year 1 US$30M; B launch US$15M)	US$26M	US$56M	B still ahead
Higher failure probability for B (50%; 70%)	n/a	US$96M; US$43M	Turns negative only near 86%
Stage gate after Year 0 (B failure stops at US$20M)	n/a	US$155M	Loss capped at US$20M
Combined downside (A bear case at 14%; B slow growth, US$20M launch revenue, 40% failure)	–US$25M	US$12M	B still NPV-positive at 11%
Break-even	Margin ≈ 6.4%; Y1 revenue ≈ US$20M	Failure probability ≈ 86%	

On a like-for-like basis, with both projects discounted at 12%, Project B's expected NPV is about US$120M against US$51M for Project A. Project B's expected NPV exceeds Project A's base case in every sensitivity we ran except three: the slow-growth case (roughly equal to Project A), the combined downside (below Project A's base case, though still positive), and a failure probability above about 67%. Figure 1 shows the spread of outcomes and Figure 2 shows cumulative cash flow. Project B's cash flows are steeper after launch, but on an expected basis they take slightly longer to turn positive.

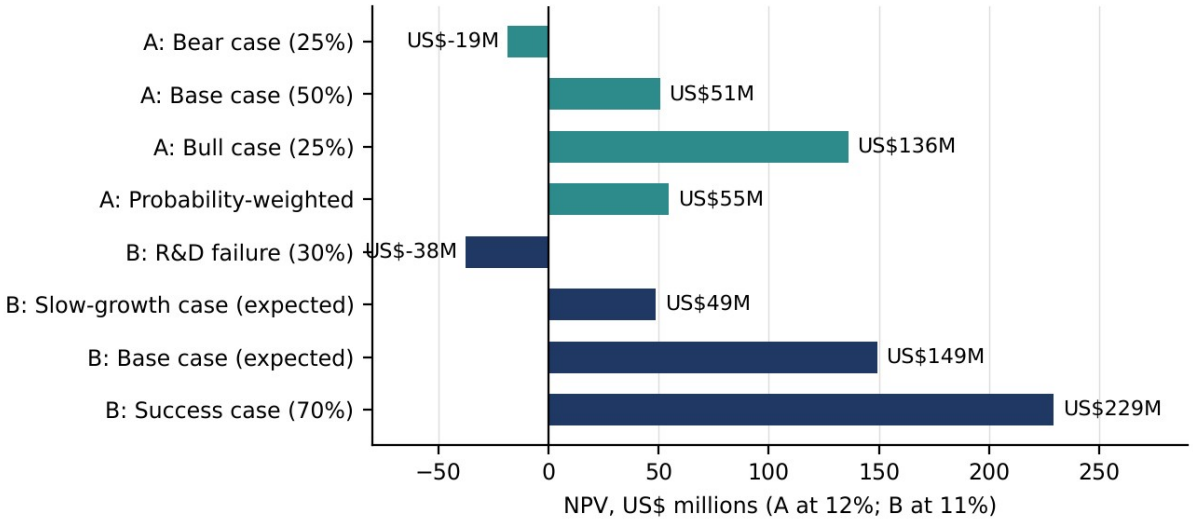


Figure 1. NPV by scenario (US$M). Project A at 12%; Project B at 11%. Failure and success cases weighted 30/70.

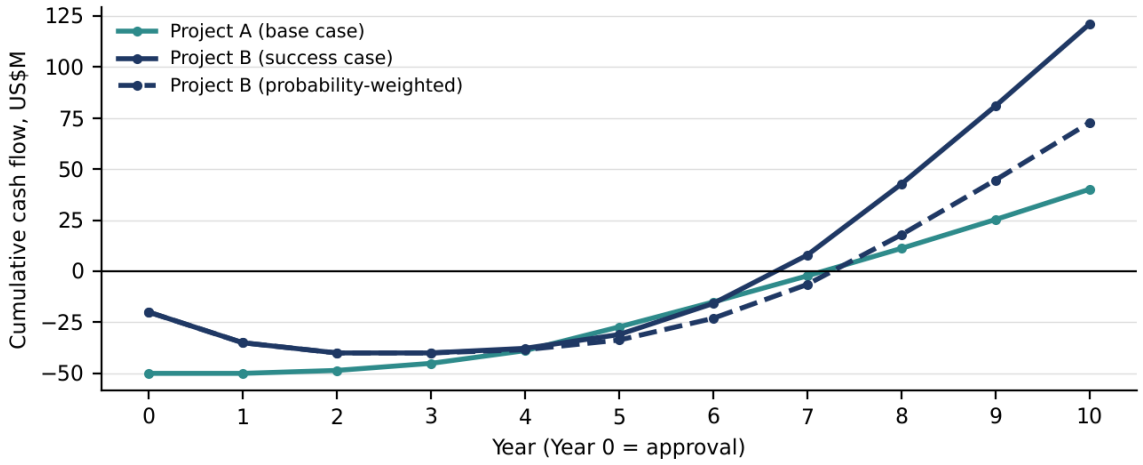


Figure 2. Cumulative undiscounted cash flow, Years 0–10 (US$M).

4. Strategic and Qualitative Assessment

Financial returns alone do not settle the decision. Table 4 compares the two projects against the factors the Board has asked us to weigh.

Table 4. Qualitative comparison

Factor	Project A (emerging markets)	Project B (disruptive R&D)	Favours
Strategic alignment	Strong fit with diversification, new customer segments and long-term growth objectives	Strong fit with maintaining market leadership and creating a competitive advantage	Even
Capability and execution	Limited prior experience in target markets; new infrastructure and local partners required	Strong internal R&D team and deep expertise in the core market; existing channels	B
Nature of risk	Mostly external and hard to control: geopolitics, regulation, FX, local competitors	Mostly internal and controllable: technical progress, IP, time-to-market; 30% failure risk	B
Reversibility	Large upfront commitment to fixed infrastructure; limited exit options	Gated: the company can stop before the bulk of the spend	B
Downside if things go wrong	Bear-case NPV negative; stranded assets and partner dependence	Failure loss capped near US$40M (about US$20M with a gate)	B

Factor	Project A (emerging markets)	Project B (disruptive R&D)	Favours
Revenue diversification	Strong: new geographies and customer segments reduce reliance on the core market	Limited: concentrated in the existing core market	A
Cost of inaction	Neutral: no immediate threat to the core business	High: if a rival commercialises disruptive technology first, the core franchise is exposed	B
Governance and compliance	Higher: anti-corruption, sanctions, data localisation, local labour and tax exposure	Moderate: IP protection, export controls on technology	B
Capital intensity	US$50M, about 50% of available cash	US$40M, about 40% of available cash	B

Long-term value creation. Both projects are back-ended: most of the value comes after Year 5. Project A's value rests on sustained growth in markets where the company has no track record. Project B's value rests on winning a technology race, and if it succeeds it creates patentable IP and licensing options beyond the base case. **Defensive value.** Project B also protects existing revenue. Its benefit is partly defensive, so it is understated in the NPV. **Diversification.** Project A is the stronger answer to the Board's diversification objective. Diversification alone does not justify the investment, however, and a smaller, staged entry into the same markets would achieve much of the strategic benefit at lower risk.

5. Recommendation: Proceed with Project B

I recommend that the Board approve Project B for the upcoming fiscal year, on the following basis:

- **Quantitative case.** Expected NPV of about US$150M against about US$50M for Project A, a higher IRR, and less capital. Project B remains NPV-positive in every combined downside we tested.
- **Qualitative case.** It uses the company's existing R&D strength and market knowledge, carries mostly controllable risk, and protects the core franchise from a competitor's disruptive move.
- **Structural case.** The programme can be stopped at defined gates. Management can therefore buy information before committing the bulk of the capital, which is valuable under genuine uncertainty.
- **Conditions of approval.** (i) Three-tranche release: US$20M at approval, US$15M after Gate 1 (end of Year 0, about month 12), and US$5M after Gate 2 (launch readiness, about month 24). (ii) Gate criteria set in advance: a technical proof of concept meeting agreed performance targets at Gate 1; a validated prototype, a freedom-to-operate opinion, and signed customer letters of intent or co-development agreements at Gate 2. (iii) Any overrun above US$40M requires fresh Board approval. (iv) Quarterly reporting to the Board on the key risk indicators in Section 6.

Why not defer Project B in favour of A. Deferring B would forgo the largest expected value available to the company and would leave the core franchise exposed to competitive moves. Deferring A, by contrast, costs little: it preserves the option, and the company can revisit it once the scoping work (market selection, partner options, FX and regulatory review) is complete and Project B's first gate has been passed.

6. Principal Risks of Project B and Mitigation

We have identified three risks as the most material to Project B's value and delivery: one financial risk tied to the R&D outcome, one financial and commercial risk tied to revenue and margin, and one operational risk tied to protecting and scaling the technology. Table 5 summarises them. Mitigation actions and contingency plans follow.

Table 5. Top three risks to Project B

#	Risk	Type	Likelihood / impact	Key risk indicator (KRI)
R1	R&D failure: the technology does not reach the performance target	Financial	Medium (30%) / High (up to US$40M write-off)	Milestone slippage; test results versus target; cost-to-complete vs budget

#	Risk	Type	Likelihood / impact	Key risk indicator (KRI)
R2	Commercial shortfall: slower adoption, competitor response and margin pressure, or technology obsolescence before launch	Financial / commercial	Medium / High (slow case NPV about US$49M)	Customer LOI conversion; win rate; price realisation vs plan; competitor launch dates
R3	IP and execution: weak patent position, loss of key R&D staff, or manufacturing scale-up delay	Operational	Medium / Medium–High	Patent filings and oppositions; attrition of key staff; pilot-line yield; time-to-scale

Risk R1: R&D failure (financial)

- **Impact.** A 30% chance of losing most or all of the US$40M, which would produce an NPV of about –US$38M in that branch (versus about +US$229M if successful).
- **Mitigation.** Use the three-gate structure (Section 5) so that money is released only after technical evidence. Obtain independent technical review at each gate. Pursue parallel development paths for the most uncertain component. Set a fixed budget with no automatic top-up, and track cost-to-complete monthly. Hold a central contingency at company level rather than inside the project budget.
- **Contingency.** If Gate 1 is failed, stop the programme with about US$20M spent. Redeploy the R&D team to adjacent applications or licensing work where possible, and sell or license the IP that remains valuable. Release the unspent US$20M and reallocate it, via the Board, to Project A's second tranche or to shareholder returns.

Risk R2: Commercial shortfall and margin compression (financial)

- **Impact.** Slower growth alone would cut the expected NPV to about US$49M. A combined downside, with slow growth, lower launch revenue and a higher failure rate, reduces it to about US$12M. The project still clears its hurdle, but the margin for error is much smaller.
- **Mitigation.** Co-develop the product with 3–5 anchor customers and secure letters of intent before Gate 2. Set a minimum price and margin floor for the launch. Keep the product architecture modular so that upgrades can be made quickly if a rival moves first. Monitor competitor roadmaps and patent filings through a standing intelligence process reviewed each quarter.
- **Contingency.** If adoption tracks the slow case, reduce go-to-market spending, shift part of the revenue to OEM or licensing partners, and phase production to match demand. If steady-state margin tracks below about 12%, run a cost-reduction programme and renegotiate supplier terms. Both responses can be executed without new capital.

Risk R3: IP protection and execution (operational)

- **Impact.** Weak or circumventable patents would erode the competitive advantage that justifies the margin assumption. Loss of key R&D staff or a slow scale-up would delay revenue and could push the launch beyond the assumed timetable.
- **Mitigation.** File priority patents in the core and key export markets early, and complete a freedom-to-operate analysis before Gate 2. Apply trade-secret controls to the core know-how. Put retention arrangements in place for the key R&D staff, tied to milestones. Document the design and knowledge base to reduce key-person dependence. Qualify a second supplier and run a pilot production line before committing to full-scale tooling.
- **Contingency.** If a patent is challenged, pursue defensive publication, cross-licensing with peers, or IP insurance as appropriate. If key staff leave, use a pre-agreed contract manufacturer or engineering partner, or acquire capability through a targeted hire. If scale-up slips, hold the Tranche 3 spend until the production line meets its yield and quality targets, and tell the Board.

7. Capital Allocation if Both Ventures Proceed

Some members have asked whether the company should pursue both projects. On our directional estimates both clear the 9% WACC, so both are viable. The question is how to allocate the US$100M. Table 6 compares three options. Our recommendation is Option C, a staged allocation.

Table 6. Allocation options

Option	Allocation	Committed at approval	Assessment
A. Full commitment now	A: US$50M; B: US$40M; reserve: US$10M	US$90M (90% of cash)	Maximises speed but leaves no gating, and a US$10M reserve is thin for an emerging-market programme
B. Project B only	B: US$40M; US$60M held	US$40M	Lowest risk and highest liquidity, but forgoes the diversification and the value of Project A
C. Staged, both projects (recommended)	A: US$50M (US$25M + US$25M); B: US$40M (US$20M + US$15M + US$5M); reserve: US$10M	US$45M (45% of cash)	Captures both projects while keeping most of the capital behind gates; the reserve covers overruns or FX

Table 7. Recommended phasing of the US$100M (Option C)

Project / bucket	Total (US$M)	Share	Release and gate conditions
Project B: Tranche 1	20	20%	At approval. Phase 1 R&D to Gate 1 (end of Year 0)
Project B: Tranche 2	15	15%	Released at Gate 1 on technical proof of concept and IP filings
Project B: Tranche 3	5	5%	Released at Gate 2 (about month 24) on validated prototype, customer LOIs and launch readiness
Project A: Tranche 1	25	25%	At approval. Entity set-up, one or two priority markets, local partner agreements and regulatory approvals
Project A: Tranche 2	25	25%	Released at a Year-2 gate on KPIs: partner performance, revenue run-rate and margin trajectory; FX hedging framework in place
Reserve (liquidity and contingency)	10	10%	Held by the CFO; used only for overruns or FX shocks, with Board notification above US$5M
Total	**100**	**100%**	**US$45M committed at approval; US$45M released only at gates**

Rationale for the staged allocation.

- **Maximum exposure is US$90M, but only US$45M is committed at approval.** If Project B fails at Gate 1, its unspent US$20M and the Year-1 and Year-2 spend are never committed. That capital can accelerate Project A or remain in reserve.
- **Project A's second tranche is conditional on evidence.** The first tranche funds the market-entry work that reduces uncertainty: partner selection, regulatory clearance and an FX framework. Only if those are in place does the company commit the rest.
- **Liquidity is protected.** The company has consistent profitability and a strong balance sheet, so operating cash flows should sustain the business. The US$10M reserve is a floor for shocks, not a substitute for cash generation. The Board should confirm a minimum liquidity policy as part of this approval.
- **The two projects diversify each other.** Project A's cash flows depend on emerging-market macro conditions and Project B's on a technology outcome. Their risks are largely uncorrelated, which reduces the chance that both

disappoint at once. On full commitment the combined expected NPV is about US$200M, of which Project A adds about US$51M for US$50M of capital.

Management bandwidth. Two concurrent programmes place real demands on leadership. Project B's first year is R&D-led, while Project A's first year is infrastructure- and partner-led, so the peaks do not coincide. We recommend separate programme offices with a shared Board-level dashboard and a single executive sponsor for portfolio risk.

8. Decisions Requested and Next Steps

The Board is asked to:

- **Approve Project B** with a US$40M envelope, releasing Tranche 1 (US$20M) now and approving the gate criteria in Section 5.
- **Defer full approval of Project A.** Direct management to complete, within 90 days, a scoping study covering market selection, partner and entry-model options (including a capped pilot of no more than US$10M), FX, regulatory and tax exposure, and an updated business case. The study should return to the Board in the next planning cycle.
- **Note the allocation framework** in Section 7 (Option C) should the Board decide to pursue both ventures, and confirm the US$10M reserve and a minimum liquidity policy.
- **Note the key assumptions** in Section 2 and direct management to validate the revenue, margin and cost inputs in the detailed business cases before any tranche is released.

Table 8. Next steps and governance

Timing	Action	Owner
0–3 months	Launch Project B Phase 1; set up the programme office, IP strategy and KRI dashboard. Complete Project A scoping study	CTO / CFO
3–12 months	Gate 1 review (proof of concept, IP filings). Release Tranche 2 if passed. Begin Project A market-entry pilot if approved	Board / Programme owners
12–24 months	Gate 2 review (prototype, LOIs, FTO opinion, launch readiness). Project A Year-2 gate if both proceed	Board
Quarterly	Board dashboard: milestones, spend vs budget, KRIs, competitor and market update	CFO

Appendix A. Annual Cash Flow Schedule (US$M, directional)

Revenue and free cash flow by year. Project B is shown as the success case and as the probability-weighted expected case (70/30). Year 0 is approval.

Table A1. Cash flows and revenue (Years 0–10)

US$M	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
A: Revenue	–	40.0	48.0	57.6	69.1	82.9	87.1	91.4	96.0	100.8	105.9
A: Free cash flow	-50.0	–	1.4	3.5	6.2	11.6	12.2	12.8	13.4	14.1	14.8
A: Cumulative FCF	-50.0	-50.0	-48.6	-45.1	-38.9	-27.3	-15.1	-2.3	11.2	25.3	40.1
B: Revenue (success)	–	–	–	30.0	45.0	67.5	101.2	131.6	157.9	173.7	182.4
B: FCF (success)	-20.0	-15.0	-5.0	–	2.2	6.8	15.2	23.7	34.7	38.2	40.1
B: FCF (failure)	-20.0	-15.0	-5.0	–	–	–	–	–	–	–	–

US$M	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
B: FCF (expected)	-20.0	-15.0	-5.0	–	1.6	4.7	10.6	16.6	24.3	26.8	28.1
B: Cumulative expected FCF	-20.0	-35.0	-40.0	-40.0	-38.4	-33.7	-23.1	-6.5	17.8	44.6	72.7

Terminal value is applied after Year 15 at 3% growth. Year 15 revenue: Project A about US$135M; Project B (success) about US$233M.

Appendix B. Methodology and Limitations

- **NPV** = sum of discounted annual free cash flows (Years 1–15) plus the discounted Gordon terminal value on Year-15 cash flow at 3% growth, with year-end timing and Year 0 as approval.
- **IRR** is the rate at which NPV equals zero, with the terminal value recomputed at each rate. For Project B, the expected IRR is calculated on probability-weighted cash flows (70% success, 30% failure).
- **Probability weighting** is applied to Project B only. For Project A, risk is captured through the discount-rate premium and the bear, base and bull scenarios (25/50/25).
- **Limitations.** The model is directional. It uses no company-specific revenue, cost or tax data, and assumes no correlation between Project A and Project B outcomes. It excludes salvage and licensing value (which would improve B's failure case), tax-structuring effects, and detailed working capital. These should be built into business cases before final approval.
- **Reproducibility.** All figures can be regenerated from the stated inputs and the cash-flow logic in Appendix A.